RTA-SUM - State of Health Manager (SOH)

Architecture Design

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1 Introduction and Goals

This document intends to describe the architecture design of the SUM_SOH module.

1.1 Introduction

This section lists some of the driving forces that software architects must consider in their decisions. In particular, we will first present the purpose of the module, and then we will describe the purpose of the document.

1.1.1 Module Purpose

This document refers to a set of standard software compositions which are named GM Standardized Utility Modules (SUM). Use of the SUM modules allows for the standardized specification of behaviorial requirements and expectations concerning the interface to the software infrastructure (BSW Services). Hence, we can completely abstract functional software from the software infrastructure. We can use SUMs to communicate a set of standardized implementations of the interactions between SW-C and BSW.

The SUM_SOH is a software module in the application layer of the standardized architecture of AUTOSAR.

SUM_SOH implements the following functionalities:

- Reset reliability monitor
- Memory Status reliability monitor
- Platform reliability monitor
- Communication reliability monitor
- Watchdog Manager reliability monitor
- Data provider via DIDs

1.1.2 Document Purpose

This document introduces the key ideas of the architecture of SUM_SOH. Its intended audience comprises System Architects, Product Architects, Developers and Testers who need a rough understanding of the concepts of the product. The document describes the product and its functionality on the component level, the interaction and communication between those components and the integration of ETAS wide shared-components. The document does not explain the design of the components in detail.

1.2 Goals of the Architecture

The main goals of the architecture are:

1. **Configurability**: The module has to be designed with the highest possible level of configurability, to be easy to adapt it in the future to changes on the standard or even in some project requirements.

2. **Maintainability**: The architecture of the component shall be clearly defined and easy to understand, to guarantee the maximum possible level of maintainability.
3. **Performance**: The component shall be designed to maximize run-time execution speed as well as minimize memory consumption.
4. **Correctness of Implementation**: The Detailed Design shall follow the requirements listed in the Architecture document (GB8002).

1.3 Stakeholders

Stakeholders for SUM_SOH module are GM, Bosch, Systems engineers, Software architect, QA personnel.

1.4 Definitions, Acronyms and Abbreviations

Term/Abbreviation	Definition
API	Application Program Interface
BSW	Basic Software
BswM	Basic Software Mode Manager
CAN	Controller Area Network
Dcm	Diagnostic Communication Module
DID	Data IDentifier
DTC	Diagnostic Trouble Code
DIAG	Diagnostic Application Services
ECU	Electronic Control Unit
EcuM	ECU State Manager
ERRH	Error Handling Module
LIN	Local Interconnect Network
LNM	LIN Network Manager
NVRAM	Non Volatile Random Access Memory
NvM	NVRAM Manager
OS	Operating System
OAW	Open Architecture Ware
PMS	Power Mode Slave
PNC	Partial Network Coordinator
RTE	Run Time Environment
SOH	State Of Health Manager
SSM	Signal Status Monitor
SSC	Security Service Coordinator
SUM	Standard Utility Module

Term/Abbreviation	Definition
SUSD	Startup Shut down Coordinator
SW	Software
SW-C	Software Component
TCS	Timing Client Service

1.5 Referenced Documents

Document Name	Description	Version	Released
SDV-8002	AUTOSAR Standard Utility Modules Specification	27.27.180.3p0	04/04/2025
SDV-6001	Diagnostic Infrastructure Specification	27.27.180.3p0	04/09/2025
CYS5402	SecOC and SOME IP Deployment Requirements	27.27.180.3p0	03/04/2025

2 Architecture Constraints

2.1 Environment

In the context of an AUTOSAR ECU, RTA-SUM modules are technically built as Application Software Components. Therefore, each RTA-SUM module requires a full AUTOSAR SW stack to run. We assume that the following AUTOSAR SW will be used:

Tool Name	Version	Description
ISOLAR-AB	12.3.3BETA1	AUTOSAR Authoring and BSW Configuration tools.
RTA-BSW	12.3.3pr1	BSW automatic configuration and code generation.
RTA-RTE	12.4.0	RTE generator used for testing
RTA-OS	6.1.0	OS that is used for testing, as AUTOSAR OS
MemMapGen	12.3.0	MemMap header file generator.
Cantata	9.5	Used for functional testing

2.2 Layers Definition

The SUM_SOH is a software module in the application layer of the standardized architecture of AUTOSAR (Figure 2.1). It interacts with other RTA-SUM modules and components of the BSW. In the latter case, the RTE is the means through which the interaction takes place.

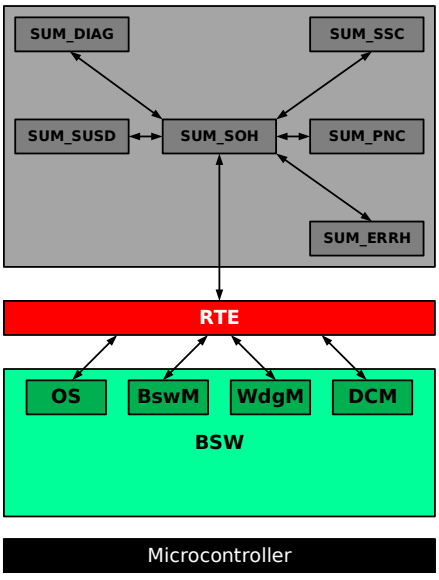


Figure 2.1: System layer overview.

2.3 Generic component interface

The external interfaces of RTA-SUM module, and most of the interactions among SUM modules are specified in GM GB 8002.

2.4 Other constraints and assumptions

The ECU in which the SUM_SOH is implemented shall have a Secure cumulative NVM memory type, as defined in CTRS section 7.1.6, accessible to the SUM_SOH. The ECU in which the SUM_SOH is implemented shall have NonClearedRAM memory type accessible to the SUM_SOH. The ECU in which the SUM_SOH is implemented shall have an AdaptableNVM memory type accessible to the SUM_SOH.

The ECU initialization sequence shall be implemented such that any NVM needed for the SUM_SOH function is available prior to the SUM_SOH being initialized.

For ECUs that implement Memory Partitioning mechanisms, the SUM_SOH shall be implemented in the supervisory partition of the ECU software. Rationale: SUM_SOH must be in the trusted partition in order to interface with OS services. Supervisory partition is expected to include BSW Modules.

The SUM_SOH shall be implemented as a Supervised Entity of the Watchdog Manager.

3 Building Block View (Static View)

The following diagram shows the internal structure of RTA-SUM SOH:

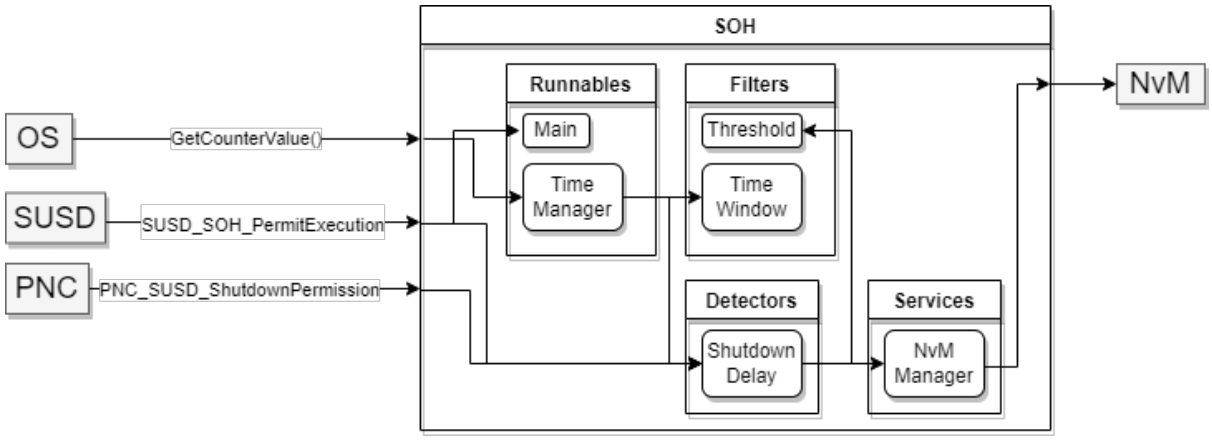


Figure 3.1: Generic Internal Module Structure

The SUM_SOH dependencies require that ECUs deploying the SUM_SOH shall also deploy the following SUMs: SUM_DIAG, SUM_SUSD, SUM_SSC, SUM_PNC and SUM_ERRH. By means of the RTE, it also interacts with the OS, BswM, DCM and WdgM (Figure 3.1).

3.1 Runnables

The Runnables are functions that start when an event occurs. They are the entry points of the feature implemented. The Runnables present in the SOH are:

- Main
- Init
- Time manager.

3.2 Filters

The SUM_SOH detectors can use one of two types of filtering:

- Threshold filtering
- Time window filtering.

3.2.1 Threshold Filter

3.2.2 Time Window Filter

[$ADS_SOH 008]
The Time Window Filter requires:

- a 2-byte unsigned integer used as detection counter to maintain a raw count of the detected events
- a 3-byte unsigned integer used as time window
- a 2-byte unsigned integer used as DID counter to maintain a filtered count of the detected events.

[$Satisfies $SRS_SOH 046] [$Satisfies $SRS_SOH 051] [$Satisfies $SRS_SOH 057]

[$ADS_SOH 009]
Each detection counter requires:

- to be stored in a specific memory type according to Table 3.1
- a counter threshold calibration value associated with it.

[$Satisfies $SRS_SOH 047] [$Satisfies $SRS_SOH 050] [$Satisfies $SRS_SOH 066]

[$ADS_SOH 010]
Each time window requires:

- to use seconds as its time base unit
- a time window threshold calibration value associated with it
- to reside in the same memory of the related detection counter.

[$Satisfies $SRS_SOH 052] [$Satisfies $SRS_SOH 053] [$Satisfies $SRS_SOH 054]

[$ADS_SOH 011]
Each DID counter requires to be stored in secure cumulative NVM.
[$Satisfies $SRS_SOH 058]

[$ADS_SOH 012]

Detection Counter Name	Memory Type
SUM_SOH_ImmeReset_Cntr	NonClearedRAM
SUM_SOH_DelayedReset_Cntr	NonClearedRAM
SUM_SOH_DiagReset_Cntr	AdaptableNVM
SUM_SOH_StatNVM_Cntr	RAM
SUM_SOH_ActNVM_Cntr	RAM
SUM_SOH_PrivNVM_Cntr	RAM
SUM_SOH_ExcepOS_Cntr	NonClearedRAM
SUM_SOH_ServOS_Cntr	NonClearedRAM
SUM_SOH_StackOS_Cntr	NonClearedRAM
SUM_SOH_MemProtOS_Cntr	NonClearedRAM
SUM_SOH_TimeProtOS_Cntr	NonClearedRAM
SUM_SOH_ErrOS_Cntr	NonClearedRAM
SUM_SOH_ShutDown_Cntr	AdaptableNVM
SUM_SOH_ClearDiagInfo_Cntr	RAM
SUM_SOH_Reject_ClearDiag_Cntr	RAM
SUM_SOH_DiagSession_Cntr	Secure Cumulative NVM
SUM_SOH_Accept_DiagSession_Cntr	Secure Cumulative NVM
SUM_SOH_SecurityAccess_Cntr	Secure Cumulative NVM
SUM_SOH_MACfail_TenAct_Cntr	RAM
SUM_SOH_ReplayCount_Cntr	RAM
SUM_SOH_SecP_Cntr	RAM
SUM_SOH_LocalStatFail_Cntr	NonClearedRAM
SUM_SOH_WdgM_DEMErr_Cntr	NonClearedRAM
ERRH_SOHSupFail_Cntr	NonClearedRAM
SOH_ERRHSupFail_Cntr	NonClearedRAM

Table 3.1: Detection Counter Memory Type

[$Satisfies $SRS_SOH 066]

3.3 Reliability monitors

The SUM_SOH supports the following reliability monitors:

- Reset Reliability monitor.
- Memory Status Reliability Monitor.
- Communication Reliability Monitor.

- Watchdog Manager Reliability Monitor.
- Platform Reliability Monitor.

3.3.1 Platform Reliability Monitor

[$ADS_SOH 001]
The scope of this monitor is to keep track of the following platform related event:

- Shutdown Delay

[$Satisfies $SRS_SOH 116]

Shutdown Delay Detector

[$ADS_SOH 020]
A Shutdown Delay is defined as a shutdown that does not occur within the ECU calibration shutdown threshold time.
[$Satisfies $SRS_SOH 117] [$Satisfies $SRS_SOH 118]

[$ADS_SOH 002]
To handle the *Shutdown Delay*, a Threshold filter is used with the following variables:

- Counter : SUM_SOH_ShutDown_Cntr
- Counter Threshold : SUM_SOH_ShutDown_Thresh
- DID : SUM_SOH_DID_ShutDown_Cntr.

These variables are stored in a specific memory type according to Subsection 3.2.2.
[$Satisfies $SRS_SOH 119] [$Satisfies $SRS_SOH 120] [$Satisfies $SRS_SOH 121] [$Satisfies $SRS_SOH 132]

[$ADS_SOH 003]
The Shutdown Delay Detector uses also an unsigned 2-byte integer timer SUM_SOH_ShutDown_Timer, to monitor the time duration that the ECU is taking to shut down. SUM_SOH_ShutDown_Timer uses seconds as its unit of time and is stored in RAM.
A flag declares if the timer SUM_SOH_ShutDown_Timer has started or not.
[$Satisfies $SRS_SOH 122] [$Satisfies $SRS_SOH 124] [$Satisfies $SRS_SOH 125]

[$ADS_SOH 021]
Moreover, a calibration value SUM_SOH_ShutDown_Timer_Thresh associated with SUM_SOH_ShutDown_Timer defines the ECU calibration shutdown threshold time.
[$Satisfies $SRS_SOH 126]

[$ADS_SOH 028]
The flow of inputs and outputs is managed thanks to the following ports:

- SUSD_SOH_PermitExecution
- PNC_SUSD_ShutdownPermission.

[$Satisfies $SRS_SOH 228] [$Satisfies $SRS_SOH 229]

3.4 DIDs

3.4.1 DID Counters

DID 0xF091: SUM_SOH Platform Reliability Monitor

[$ADS_SOH 022]
SUM_SOH makes use of DID 0xF091 to maintain the results of Platform Reliability events thanks to the following parameters:

- SUM_SOH_DID_ShutDown_Cntr : a 2-byte unsigned integer used to maintain a filtered count of the occurences of shutdown delays.

[$Satisfies $SRS_SOH 036]

3.5 Port Interfaces

All ECU types shall support Client / Server port interfaces. All ECU types shall support Sender / Receiver port interfaces.

Port Interface	Port Type	P-Port	R-Port	Description
ERRH_E_BswEvent	Sender / Receiver	SUM_ERRH	SUM_SOH	Reception of DEM events which are routed from SUM_ERRH to the SUM_SOH for processing.
SUSD_SOH_PermitExecution	Sender / Receiver	SUM_SUSD	SUM_SOH	Permit Execution interface to indicate that operating conditions are correct for the SUM_SOH to execute its functions.
PNC_SUSD_ShutdownPermission	Sender / Receiver	SUM_PNC	SUM_SOH	Notification to the SUM_SOH of any active or not active partial networks.
SSC_E_SecureMessageMonitoringEvents	Sender / Receiver	SUM_SSC	SUM_SOH	Notification of MAC failures along with their types.

Port Interface	Port Type	P-Port	R-Port	Description
DIAG_ECUResetRequest	Sender / Re-ceiver	SUM_DIAG	SUM_SOH	Notification of a service 0x11 tester-initiated request for ECU Reset (not programming).
SUSD_SOH_ResetOccured	Sender / Re-ceiver	SUM_SUSD	SUM_SOH	Notification to the SUM_SOH that a reset occurred. The type of reset is also communicated to the SUM_SOH. Type of resets may be delayed or immediate.
SSFM_ClearDiagReq	Sender / Re-ceiver	SUM_SSFM	SUM_SOH	Notification to the SUM_SOH that a service 0x04 or a service 0x14 request was accepted or rejected.
DIAG_SOH_DiagSessionRequest	Sender / Re-ceiver	SUM_DIAG	SUM_SOH	Notification to the SUM_SOH that a reprogramming request took place (service 0x10 subfunction 0x02).
SSC_SOH_SecurityAccessError	Sender / Re-ceiver	SUM_SSC	SUM_SOH	Notification to the SUM_SOH of the occurrences of service 0x27 failures.
BatteryVoltage_DisableNVMWrite	Sender / Re-ceiver	SUM_SUSD	SUM_SOH	Notification to the SUM_SOH from SUM_SUSD that vehicle battery voltage is too low to perform NVM writes.

Port Interface	Port Type	P-Port	R-Port	Description
DIAG_GetMEC	Sender / Receiver	SUM_DIAG	SUM_SOH	This interface is used by the SUM_DIAG to communicate to SUM_SOH to value of the MEC.
DataServices<SOH>Data<n>	Client / Server	DCM	SUM_SOH	To support the collection of Diagnostic DIDs that are part of the SUM_SOH functionalities.
DiagnosticReprogrammingRequest	Sender / Receiver	BswM	SUM_SOH	Notification of a request to shutdown to reset as part of programming to jump to boot.
SOH_WdgM_IndividualMode<1...n>	Mode Switch	WdgM	SUM_SOH	Notification of WdgM local supervision failures occurrence for all supervised entities excluding the SUM_SOH and the SUM_ERRH. These interfaces replace the SOH_BswM_WdgM_IndividualMode<1...n> interfaces described in GB8002.
SOH_WdgM_IndividualMode_Errh	Mode Switch	WdgM	SUM_SOH	Notification of the WdgM SUM_ERRH local supervision failures occurrences. This interface replaces the SOH_BsWM_WdgM_ERRHMode interface described in GB8002.

Port Interface	Port Type	P-Port	R-Port	Description
RP_ErrorStatus	Mode Switch	BswM	SOH	This interface is used by the BswM to notify SUM_SOH of SecP errors occurrences.
NvMService	Client / Server	NvM	SOH	The interface provides services for the management and maintenance of NV data related to SOH Metrics.
TCS_SWC_Clock	Client / Server	TCS	SOH	Provides Global Time for timestamp (unit 100ms).
RoutineServices_Clear_SUM_SOH_logs	Client / Server	DCM	SOH	The interface provides services for the clear of the SOH Metrics.
WdgM_LocalSupervision	Client / Server	WdgM	SUM_SOH	Notify the checkpoint reachement for implementing the Local Supervision.
SOH_ERRH_ClearRID0x02C0	Client / Server	SUM_ERRH	SUM_SOH	The interface provides service for starting clear, all the metrics related to SUM_SOH Local Supervision Detector implemented at ERRH side.

Port Interface	Port Type	P-Port	R-Port	Description
SOH_SUSD_ShutdownPermission	Sender / Receiver	SUM_SOH	SUM_SUSD	Shutdown Permission interface to indicate that operating conditions are correct for the SUM_SOH to execute the shutdown.

3.6 Configuration

SUM_SOH is a SWC within the context of AUTOSAR. It consists of static files and generated files. SUM_SOH configuration is done the same way as for standard BSW modules. A parameter definition file is included with the module which allows users to configure the available parameters. The generated files change based on the configuration.

For default values of calibration thresholds SOH is referring to the SDV-8002. Configuration parameters shall be pre-compile, in other words the ECU behavior modification requires a reflash of the complete application software.

3.6.1 Included Containers

Name	Multiplicity	Scope / Dependency
SOHGeneral	1..1	General configuration parameters of the SOH
SUSDConfigSet	1..1	This container contains the configuration parameters and sub containers of the SOH module supporting multiple SWC configuration sets

3.6.2 SOHGeneral

Parameter Name	SOH_PermitExecution_Mask
Description	Mask that can be OR'd with the value supplied by PermitExecution to set any "don't care" bits to '1'.
Multiplicity (upper..lower)	1..1
Category	ECUC-INTEGER-PARAM-DEF
Range	0..255
Default value	190

Parameter Name	SOH_RunnableSOHMainPeriod
Description	Time in sec, for the periodicity of the Runnable_SOH_Main.
Multiplicity (upper..lower)	1..1
Category	ECUC-INTEGER-PARAM-DEF
Range	0.001..0.2
Default value	0.1

Parameter Name	SohGenerateInitRunnable
Description	This parameter, when configured to TRUE, generates a periodic Runnable Entity that is useful for calling the SOH_Init() function just once.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	0

Parameter Name	SohInitFuncPeriodicity
Description	This parameter is used to specify the time in ms for the periodicity of the runnable Runnable_SOH_Init.
Multiplicity (upper..lower)	0..1
Category	ECUC-FLOAT-PARAM-DEF
Range	1.0...5000.0
Default value	1000.0

Parameter Name	SOH_StatNvM_SupportEnabled
Description	This parameter is used to enable the Nvm Status Failure/Integrity Error class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	1

Parameter Name	SOH_ActNvM_SupportEnabled
Description	This parameter is used to enable the Nvm Action Failure Error class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	1

Parameter Name	SOH_PrivNvM_SupportEnabled
Description	This parameter is used to enable the Nvm Privilege class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	1

Parameter Name	SOH_ExcepOS_SupportEnabled
Description	This parameter is used to enable the OS Hardware Protection Action Failure Error class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	true

Parameter Name	SOH_ServOS_SupportEnabled
Description	This parameter is used to enable the OS Service Protection / Trusted Function Action Failure Error class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	true

Parameter Name	SOH_StackOS_SupportEnabled
Description	This parameter is used to enable the OS Stack Fault class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	true

Parameter Name	SOH_MemProtOS_SupportEnabled
Description	This parameter is used to enable the OS Memory Protection Failure Error class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	true

Parameter Name	SOH_TimeProtOS_SupportEnabled
Description	This parameter is used to enable the OS Time Protection Action Failure Error class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	true

Parameter Name	SOH_ErrOS_SupportEnabled
Description	This parameter is used to enable the OS Other OS Errors Action Failure Error class and the relative detector.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	true

Parameter Name	SOH_RID_02C0_SOH_Clear_Log_Reference
Description	Reference to DCM RID 0x02C0 SOH Clear Logs
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

3.6.3 SOHConfigSet

SOH_ResetReliabilityMonitor

This container includes the configuration parameters needed for the Reset Reliability Monitor detectors.

SOH_ImmediateReset

Parameter Name	SUM_SOH_ImmeReset_Thresh_Init
Description	Initial value of SUM_SOH_ImmeReset_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_ImmeReset_TW_Thresh_Init
Description	Initial value of SUM_SOH_ImmeReset_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMImmResetDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F08F_ImmResetData_Reference
Description	Reference to DCM ImmReset Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_DelayedReset

Parameter Name	SUM_SOH_DelayedReset_Thresh_Init
Description	Initial value of SUM_SOH_DelayedReset_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_DelayedReset_TW_Thresh_Init
Description	Initial value of SUM_SOH_DelayedReset_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMDelayedResetDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F08F_DelayedResetData_Reference
Description	Reference to DCM DelayedReset Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_DiagnosticReset

Parameter Name	SUM_SOH_DiagReset_Thresh_Init
Description	Initial value SUM_SOH_DiagReset_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SOH_NvMDiagResetCntr_Reference
Description	Reference to the NVM Counter Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_NvMDiagResetDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F08F_DiagResetData_Reference
Description	Reference to DCM DiagReset Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_MemoryStatusReliabilityMonitor

This container includes the configuration parameters needed for the Memory Status Monitor detectors.

SOH_NVMErrorsDetector
This container includes the configuration containers needed for the NVM Error detectors.

SOH_StatusNVM

Parameter Name	SUM_SOH_StatNVM_Thresh_Init
Description	Initial value of SUM_SOH_StatNVM_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_StatNVM_TW_Thresh_Init
Description	Initial value of SUM_SOH_StatNVM_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMStatNVMDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_StatusNVM_Reference
Description	Reference to DCM StatusNVM Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_ActNVM

Parameter Name	SUM_SOH_ActNVM_Thresh_Init
Description	Initial value of SUM_SOH_ActNVM_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_ActNVM_TW_Thresh_Init
Description	Initial value of SUM_SOH_ActNVM_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMActNVMDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_ActNVM_Reference
Description	Reference to DCM ActNVM Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_PrivNVM

Parameter Name	SUM_SOH_PrivNVM_Thresh_Init
Description	Initial value of SUM_SOH_PrivNVM_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_PrivNVM_TW_Thresh_Init
Description	Initial value of SUM_SOH_PrivNVM_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMPrivNVMDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_PrivNVM_Reference
Description	Reference to DCM PrivNVM Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_OSErrorsDetector
The following configuration parameters are used to set the initial threshold value for each OS error class detection counter and their related time windows.

SOH_ExcepOS

Parameter Name	SUM_SOH_ExcepOS_Thresh_Init
Description	Initial value of SUM_SOH_ExcepOS_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_ExcepOS_TW_Thresh_Init
Description	Initial value of SUM_SOH_ExcepOS_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMExcepOSDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_ExcepOS_Reference
Description	Reference to DCM ExcepOS Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_ServOS

Parameter Name	SUM_SOH_ServOS_Thresh_Init
Description	Initial value of SUM_SOH_ServOS_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_ServOS_TW_Thresh_Init
Description	Initial value of SUM_SOH_ServOS_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMServOSDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_ServOS_Reference
Description	Reference to DCM ServOS Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_StackOS

Parameter Name	SUM_SOH_StackOS_Thresh_Init
Description	Initial value of SUM_SOH_StackOS_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_StackOS_TW_Thresh_Init
Description	Initial value of SUM_SOH_StackOS_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMStackOSDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_StackOS_Reference
Description	Reference to DCM StackOS Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_MemProtOS

Parameter Name	SUM_SOH_MemProtOS_Thresh_Init
Description	Initial value of SUM_SOH_MemProtOS_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_MemProtOS_TW_Thresh_Init
Description	Initial value of SUM_SOH_MemProtOS_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMMemProtOSDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_MemProtOS_Reference
Description	Reference to DCM MemProtOS Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_TimeProtOS

Parameter Name	SUM_SOH_TimeProtOS_Thresh_Init
Description	Initial value of SUM_SOH_TimeProtOS_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_TimeProtOS_TW_Thresh_Init
Description	Initial value of SUM_SOH_TimeProtOS_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMTimeProtOSDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_TimeProtOS_Reference
Description	Reference to DCM TimeProtOS Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_ErrOS

Parameter Name	SUM_SOH_ErrOS_Thresh_Init
Description	Initial value of SUM_SOH_ErrOS_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_ErrOS_TW_Thresh_Init
Description	Initial value of SUM_SOH_ErrOS_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMErrOSDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F090_ErrOS_Reference
Description	Reference to DCM ErrOS Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SUM_SOH_ShutDown_Thresh_Init
Description	Initial value of SUM_SOH_ShutDown_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_ShutDown_Timer_Thresh_Init
Description	Initial value of SUM_SOH_ShutDown_Timer_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	60

SOH_CommunicationReliabilityMonitor

This container includes the configuration parameters needed for the Communication Reliability Monitor detectors. Here organized by events.

SOH_ClearDiagInfo

Parameter Name	SUM_SOH_ClearDiagInfo_Thresh_Init
Description	Initial value of SUM_SOH_ClearDiagInfo_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SOH_NvMClearDiagInfoDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F092_ClearDiagInfoDID_Reference
Description	Reference to DCM Clear Diag Infomation Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_RejectClearDiag

Parameter Name	SUM_SOH_Reject_ClearDiag_Thresh_Init
Description	Initial value of SUM_SOH_Reject_ClearDiag_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SOH_NvMRejectClearDiagDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F092_RejectClearDiagDID_Reference
Description	Reference to DCM Reject Clear Diag Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_DiagSession

Parameter Name	SUM_SOH_DiagSession_Thresh_Init
Description	Initial value of SUM_SOH_DiagSession_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SOH_NvMDiagSessionCntr_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_NvMDiagSessionDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F092_DiagSessionDID_Reference
Description	Reference to DCM Diag Session Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_Accept_DiagSession

Parameter Name	SUM_SOH_Accept_DiagSession_Thresh_Init
Description	Initial value of SUM_SOH_Accept_DiagSession_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SOH_NvMAccept_DiagSessionCntr_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_NvMAccept_DiagSessionDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F092_Accept_DiagSessionDID_Reference
Description	Reference to DCM Accept Diag Session Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_SecurityAccess

Parameter Name	SUM_SOH_SecurityAccess_Thresh_Init
Description	Initial value of SUM_SOH_SecurityAccess_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SOH_NvMSecurityAccessCntr_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_NvMSecurityAccessDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F092_SecurityAccessDID_Reference
Description	Reference to DCM Security Access Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_MACfailTenAct

Parameter Name	SUM_SOH_MACfail_TenAct_Thresh_Init
Description	Initial value of SUM_SOH_MACfail_TenAct_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	$2\times$ number of received MAC protected messages

Parameter Name	SUM_SOH_MACfail_TenAct_TW_Thresh_Init
Description	Initial value of SUM_SOH_MACfail_TenAct_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	2

Parameter Name	SOH_NvMMACfail_TenActDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F092_MACfail_TenActDID_Reference
Description	Reference to DCM MAC fail type 1 and 2 Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_ReplayCount

Parameter Name	SUM_SOH_ReplayCount_Thresh_Init
Description	Initial value of SUM_SOH_ReplayCount_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_ReplayCount_TW_Thresh_Init
Description	Initial value of SUM_SOH_ReplayCount_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SUM_SOH_ReplayCount_Start_Thresh_Init
Description	Initial value of SUM_SOH_ReplayCount_Start_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	5

Parameter Name	SOH_NvMReplayCountDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F092_ReplayCountDID_Reference
Description	Reference to DCM MAC fail type 3 Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_SecP

Parameter Name	SUM_SOH_SecP_Thresh_Init
Description	Initial value of SUM_SOH_ReplayCount_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_SecP_TW_Thresh_Init
Description	Initial value of SUM_SOH_ReplayCount_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	3600

Parameter Name	SOH_NvMSecPDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F092_SecPDID_Reference
Description	Reference to DCM SecP Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_WatchdogManagerReliabilityMonitor

This container includes the configuration parameters needed for the Watchdog Manager Reliability Monitor detectors.

Local_Supervision_Failures_Detector

Parameter Name	SUM_SOH_ImmeReset_Thresh_Init
Description	Initial value of SUM_SOH_LocalStatFail_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_LocalStatFail_TW_Thresh_Init
Description	Initial value of SUM_SOH_LocalStatFail_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	10

Parameter Name	Supervised_Entity_Reference
Description	Reference to a Supervised Entity container in the WdgM
Multiplicity (upper..lower)	0..Infinite
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_LocalSupervisionFailures_DID_counter_Reference
Description	Reference to the NVM Block related to SUM_SOH_DID_LocalStatFail_Cntr
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F093_LocalStatFail_Reference
Description	Reference to DCM LocalStatFail Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

WdgM_DEM_Supervision_Errors_Detector

Parameter Name	SUM_SOH_WdgM_DEMErr_Thresh_Init
Description	Initial value of SUM_SOH_WdgM_DEMErr_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_WdgM_DEMErr_TW_Thresh_Init
Description	Initial value of SUM_SOH_WdgM_DEMErr_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	10

Parameter Name	SOH_WdgMDEMErr_DID_counter_Reference
Description	Reference to the NVM Block related to SUM_SOH_DID_WdgM_DEMErr_Cntr
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F093_WdgMDEMErr_Reference
Description	Reference to DCM WdgMDEMErr Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SUM_ERRH_Local_Supervision_Detector

Parameter Name	SOH_ERRHSupFail_Thresh_Init
Description	Initial value of SOH_ERRHSupFail_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SOH_ERRHSupFail_TW_Thresh_Init
Description	Initial value of SOH_ERRHSupFail_TW_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	10

Parameter Name	SOH_ERRHLocalSupervisionFailures_DID_counter_Reference
Description	Reference to the NVM Block related to SUM_SOH_DID_ERRHSupFail_Cntr
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F093_ERRHSupFail_Reference
Description	Reference to DCM ERRHSupFail Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F093_ERRH_Local_Supervision_Status_Reference
Description	Reference to DCM ERRH Local Supervision Status Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

SOH_PlatformReliabilityMonitor

This container includes the configuration parameters needed for the Platform Reliability Monitor detectors. Here organized by events.

SOH_ShutDownDelay

Parameter Name	SUM_SOH_Shutdown_Thresh_Init
Description	Initial value of SUM_SOH_Shutdown_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	1

Parameter Name	SUM_SOH_ShutDown_Timer_Thresh_Init
Description	Initial value of SUM_SOH_ShutDown_Timer_Thresh
Multiplicity (upper..lower)	1..1
Category	–
Range	-..-
Default value	60

Parameter Name	SOH_NvMShutdownCntr_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_NvMShutdownDID_Reference
Description	Reference to the NVM DID Block
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

Parameter Name	SOH_DID_F091_ShutdownDID_Reference
Description	Reference to DCM Shutdown DID counter Data
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	-

4 Runtime View (Dynamic View)

4.1 Runnables

4.1.1 Init

This periodic runnable is responsible for calling the SOH_init function once in order to load the values stored in the NvM and initialize the variables to their default value. In addition, it is necessary to specify that the runnable is generated, hence available, only if the `SohGenerateInitRunnable` parameter is configured to True.

4.1.2 Main

This periodic runnable is responsible of reading the SUSD_SOH_PermitExecution port and passing it to the Shutdown Delay Detector.

4.1.3 Time manager

This periodic runnable calls the OS API GetCounterValue() to read the counter configured as reference in SOHGeneral/Sum_SOH_Os_Counter_Ref and pass it to:

- the Time Window Filter
- the ShutDown Delay Detector.

The periodicity of this runnable is configurable.

The following diagram shows the Time Manager Runnable workflow.

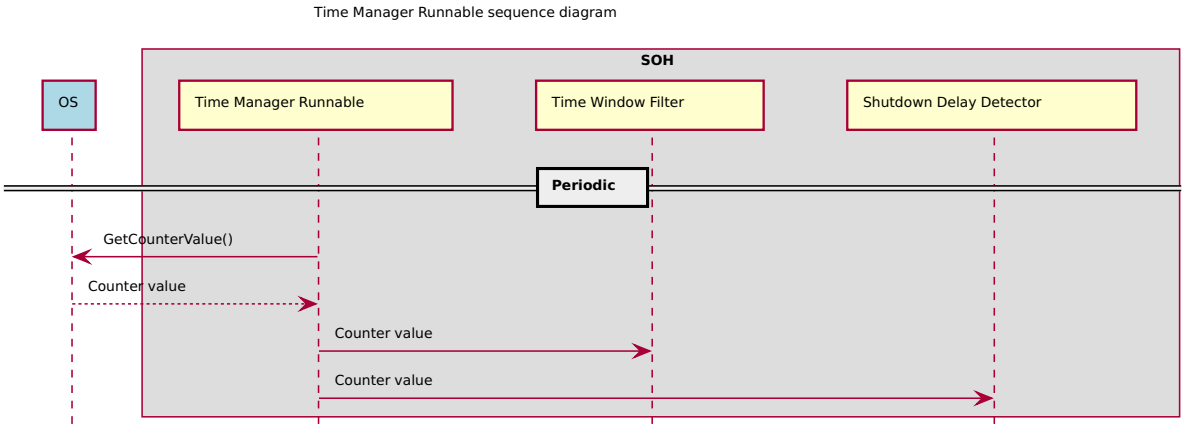


Diagram 4.1: Time Manager Runnable workflow diagram

4.2 Filters

4.2.1 Threshold Filter

4.2.2 Time Window Filter workflow

[$ADS_SOH 013]
At ECU production, each detection counter, time window and DID counter is initialized to a value of 0x0000.
[$Satisfies $SRS_SOH 048] [$Satisfies $SRS_SOH 055] [$Satisfies $SRS_SOH 059]

[$ADS_SOH 014]
At the detection of the first error occurrence in a monitoring cycle, SUM_SOH:

- starts its time window associated to that detector
- continues counting error conditions in its detection counter.

[$Satisfies $SRS_SOH 062]

[$ADS_SOH 015]
If the detection counter reaches its threshold count calibration within the calibration time window, SUM_SOH:

- increments the DID counter by one
- resets the detection counter to 0x0000 and resets its time window to 0x00000 to start a new detection cycle.

[$Satisfies $SRS_SOH 063]

[$ADS_SOH 016]
Instead, if the detection counter does not reach its threshold count calibration within the calibration time window, SUM_SOH only resets its detection counter to 0x0000 and resets its time window to 0x000000 to start a new detection cycle.
[$Satisfies $SRS_SOH 064]

[$ADS_SOH 017]
Once a new detection cycle is started, the detector waits for the occurrence of the first error event detection to start its time window.
[$Satisfies $SRS_SOH 065]

[$ADS_SOH 018]
If DID counter reaches its maximum value 0xFFFE, it is retained until the next execution of RID 0x02C0 Clear_SUM_SOH_logs.
[$Satisfies $SRS_SOH 061]

[$ADS_SOH 019]
During execution of RID 0x02C0 Clear_SUM_SOH_logs, each detection counter, time window and DID counter is cleared to a value of 0x0000.
[$Satisfies $SRS_SOH 049] [$Satisfies $SRS_SOH 056] [$Satisfies $SRS_SOH 060]

[$ADS_SOH 025]
The calculation of the elapsed time is based on the ticks of a counter read periodically by the Time Manager Runnable (see 4.1.3).
The difference between the current ticks value and the previous one is converted in seconds: at least 1 second should elapse to increase a time window.
The previous value is initialized to 0 and reset to 0 when a new detection cycle starts.
If the previous value is 0, the difference is not computed.
[$Satisfies $SRS_SOH 051] [$Satisfies $SRS_SOH 052]

The following diagram shows the Time Window Filter workflow.

Time Window Filter sequence diagram

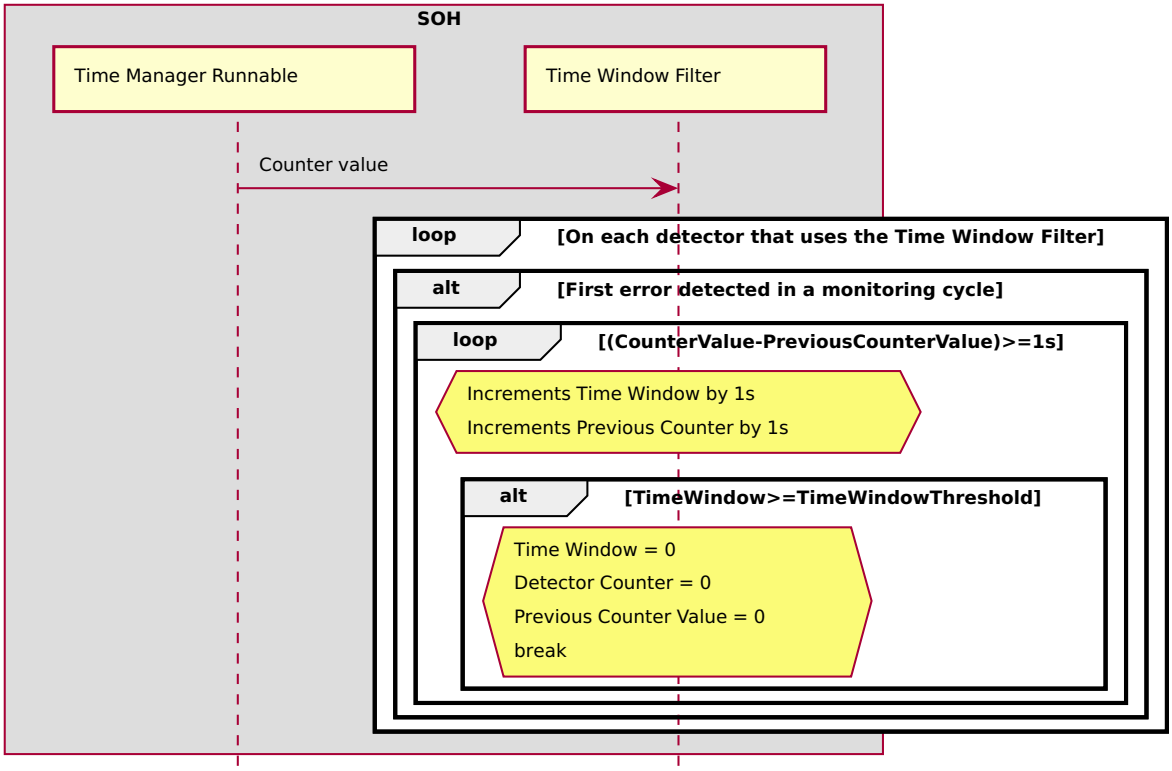


Diagram 4.2: Time Window Filter workflow diagram

4.3 Reliability Monitors

4.3.1 Platform Reliability Monitor

ShutDown Delay Detector Workflow

[$ADS_SOH 004]
The SUM_SOH_ShutDown_Timer is initialized to a value of 0x0000 at ECU production.
[$Satisfies $SRS_SOH 123]

[$ADS_SOH 005]
SUM_SOH_ShutDown_Timer starts when all the following conditions are met:

- PNC indication that all partial networks were deactivated is received by SOH through the PNC_SUSD_ShutdownPermission port
- The ECU shutdown target bit in the SUSD_SOH_PermitExecution port is set to OFF (bit 1 has value 0).

If the condition are met, the timer SUM_SOH_ShutDown_Timer's flag is set.
[$Satisfies $SRS_SOH 127] [$Satisfies $SRS_SOH 231]

[$ADS_SOH 006]
The SUM_SOH increments its timer SUM_SOH_ShutDown_Timer until it receives a callback from BswM indicating that the shutdown action list has been invoked.
The callback is scheduled in order to complete all the operations that involve the NvM.
[$Satisfies $SRS_SOH 128]

[$ADS_SOH 007]
In the case where a wakeup event happens during shutdown phase and the shutdown is aborted, the SUM_SOH is notified with the least significant bit in the SUSD_SOH_PermitExecution port equals to 1.
[$Satisfies $SRS_SOH 129]

[$ADS_SOH 023]
Upon being notified of a wakeup event occurrence while monitoring for the shutdown delay, the SUM_SOH clears its timer SUM_SOH_ShutDown_Timer.
[$Satisfies $SRS_SOH 130]

[$ADS_SOH 024]
If SUM_SOH_ShutDown_Timer reaches its threshold value, the detection counter SUM_SOH_ShutDown_Cntr is incremented by 1.
[$Satisfies $SRS_SOH 131]

[$ADS_SOH 026]
Once the counter is incremented, the filter associated with this Detector - the Threshold Filter - runs.
[$Satisfies $SRS_SOH 119]

[$ADS_SOH 027]
As for the Time Window Filter (see 4.2.2), the calculation of the elapsed time is based on the ticks of a counter read

- periodically by the Time Manager Runnable and
- when the shutdown action list has been invoked.

The difference between the current ticks value and the previous one is converted in seconds: at least 1 second should elapse to increase SUM_SOH_ShutDown_Timer.
If the previous value has not been initialized yet, the difference is not computed.
[$Satisfies $SRS_SOH 117] [$Satisfies $SRS_SOH 124] [$Satisfies $SRS_SOH 128]

The following diagram shows the Shutdown detection workflow.

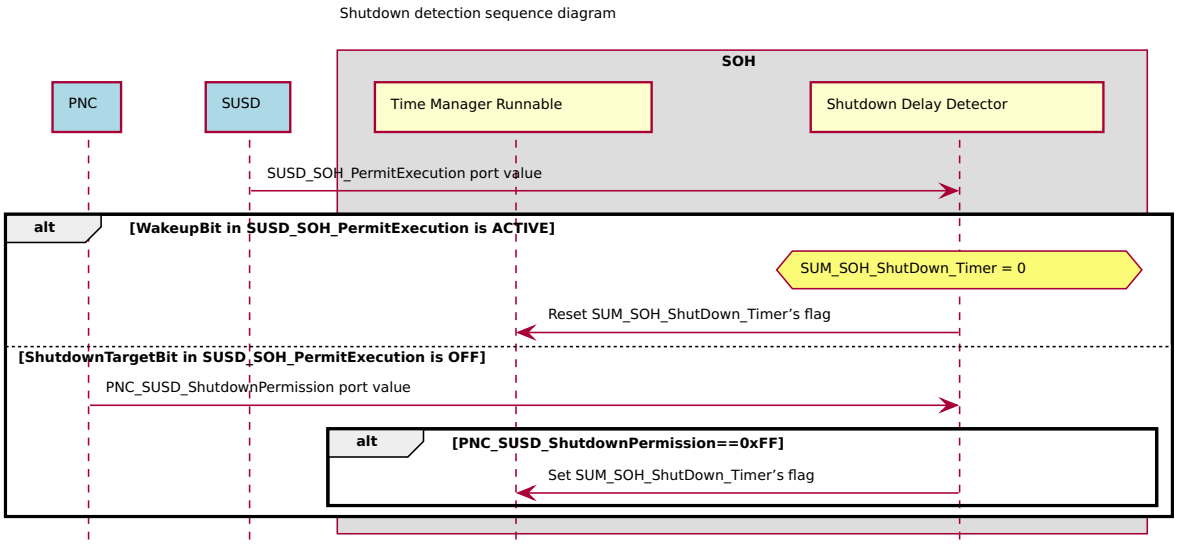


Diagram 4.3: Shutdown detection diagram

Instead, the following diagram shows the Shutdown Delay detection workflow.

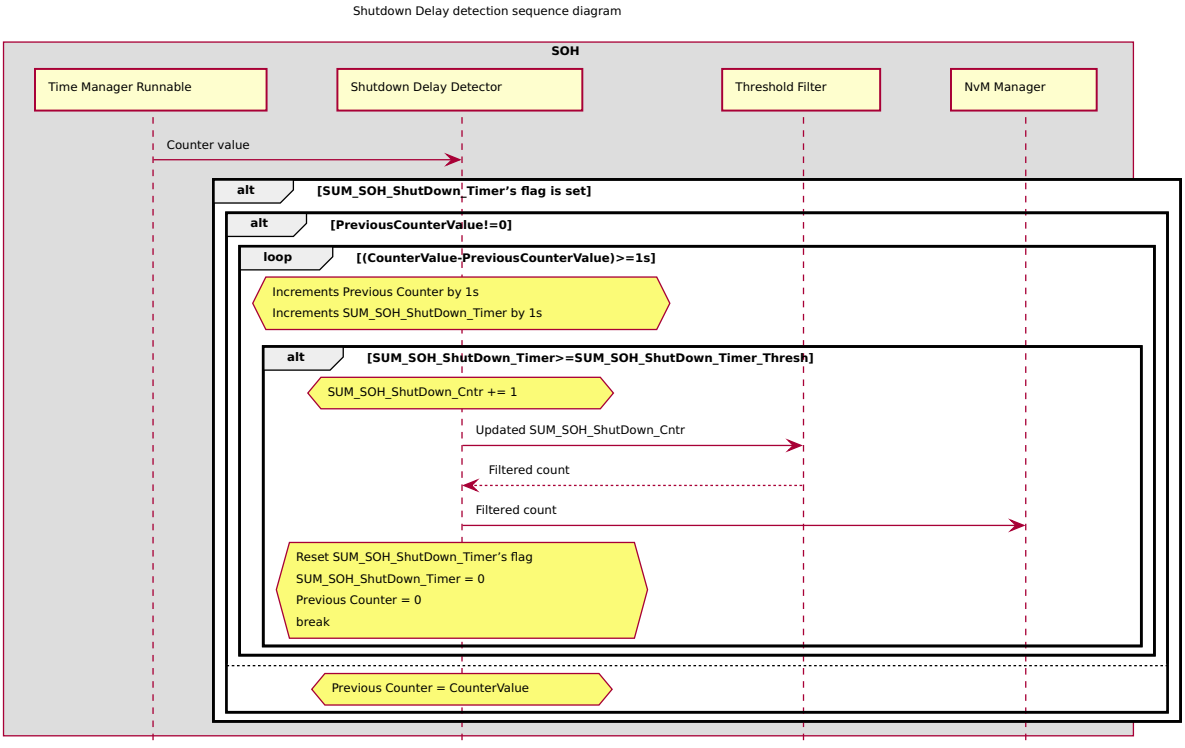


Diagram 4.4: Shutdown Delay detection diagram

5 Deployment View

5.1 Resource Limits

N/A

5.2 AUTOSAR Resources

N/A

5.3 Interrupts

N/A

5.4 Watchdog

N/A

5.5 External hardware dependencies

N/A

6 Testing Strategy

6.1 Time Window Filtering - initialization phase

N/A

6.2 Shutdown Delay timer - Shutdown phase

N/A